

Speed Controller: Standard Type Elbow Type (Metal Body)

AS Series

RoHS

Minimizes installation time and cost

Fittings and tubing are not necessary because this type screws directly into the actuator. Thus, piping labor and cost can be eliminated.

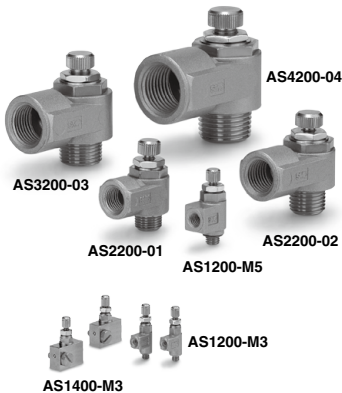
Body swivels 360°

Swivel type allows free setting of piping.

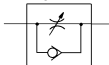
Speed may be accurately controlled even at low speeds.

Able to control and set a constant speed easily.

Retainer prevents accidental loss of needle.



Symbol



Made to Order (For details, refer to page 893.)

Symbol	Specifications
-X12	Lubricant: Vaseline
-X21	Grease-free (Seal: Fluorine-coated) + Throttle valve (Without check valve)
-X214	Throttle valve (Without check valve)

Specifications

Specifications	Model	AS1200-M3	AS1400-M3	AS12□0-M5	AS12□0-U10/32	AS22□0-01	AS22□0-02	AS32□0-03	AS42□0-04
Port size		M3 x 0.5	M5 x 0.8	10-32 UNF	1/8	1/4	3/8	1/2	
Fluid		Air							
Proof pressure		1.05 MPa		1.5 MPa					
Max. operating pressure		0.7 MPa		1 MPa					
Min. operating pressure		0.1 MPa		0.1 MPa					
Ambient and fluid temperature		-5 to 60°C (No freezing)							
Option		Round lock nut				With seal, Round lock nut, Nickel plated			
Weight (g)		3	6	10	29	64	106	181	
Controlled flow	Flow rate L/min (ANR)	20		105	230	460	920	1700	
Free flow	Sonic conductance dm ³ /s/bar	0.06		0.32	0.7	1.4	2.8	5.2	
Critical pressure ratio	Controlled flow	0.2		0.2	0.25	0.3	0.25	0.25	
	Free flow	0.4		0.4	0.2	0.3	0.2	0.3	

Note 1) Flow rate values are measured at 0.5 MPa and 20°C.

Note 2) Meter-in type not available on AS1200-M3, AS1400-M3.

Note 3) Distinction between meter-out/meter-in types by appearance.

Those are distinguished by the lock nut. The meter-out type is zinc chromated (the round lock nut is electroless nickel plated), while the meter-in type is black zinc chromate plated.

Note 4) AS12□0, AS1400, AS22□0, AS32□0: Brass parts are electroless nickel plated, provided as standard. (N specifications)

How to Order

AS 2 2 0 0 - 01 - S - - -

Body size

1	M3, M5 standard
2	1/8, 1/4 standard
3	3/8 standard
4	1/2 standard

Type

2	Direct cylinder elbow type
4	Direct cylinder flat elbow type

Control type

0	Meter-out
1	Meter-in

Thread type

Symbol	Cylinder side	Tube side
NII	R	Rc
	Metric thread (M3, M5)	
	Unified thread (10-32 UNF)	
N	NPT	
F	R *1	G *2

*1 When using a G female thread, be sure to select the option with a seal (S) as the thread requires sealant. Also, if a G thread is required for on the cylinder side, it will need to be special ordered.

*2 The R thread can be screwed into the G female thread.

Made to Order

Refer to page 893 for details.

Lock nut option

NII	Hexagon lock nut
J	Round lock nut

Option

NII	None
S	With seal
N	Electroless nickel plated (AS42□0 only)

* If more than one option is required, write option part numbers in the order of "S", "N".

Note) M3 and M5 sizes: s (with seal) is not necessary.

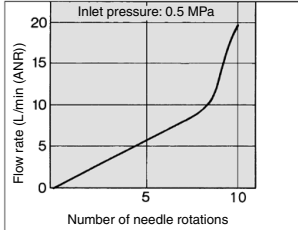
Bore size

Symbol	Bore size	Applicable series
M3	M3 x 0.5	AS1200-M3 AS1400-M3
M5	M5 x 0.8	AS12□0-M5
U10/32	10-32 UNF	AS12□0-U10/32
01	1/8	AS22□0-01
02	1/4	AS22□0-02
03	3/8	AS32□0-03
04	1/2	AS42□0-04

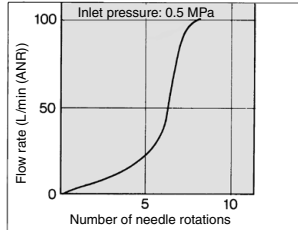
Needle Valve/Flow Rate Characteristics

Note) The flow rate characteristics are representative values.

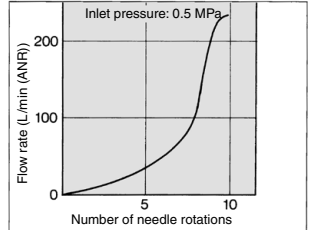
AS1200-M3, AS1400-M3



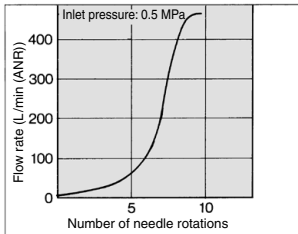
AS12□0-M5, AS12□0-U10/32



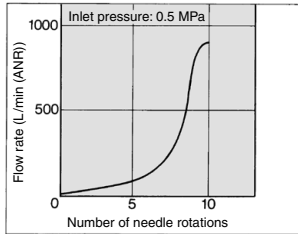
AS22□0-01



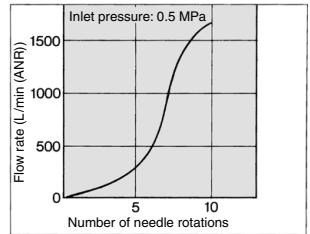
AS22□0-02



AS32□0



AS42□0



⚠ Caution

Be sure to read this before handling the products.
Refer to page 11 for safety instructions and pages 19 to 22 for flow control equipment precautions.

Made to Order



Lubricant: Vaseline

X12

Grease-free (Seal: Fluorine Coated) +
Throttle Valve (Without Check Valve)

X21

Ex.) AS2200-01-X12

Ex.) AS2200-01-X21

Note 1) Not particle-free

Note 2) This product is a restrictor without a check valve (no control direction).

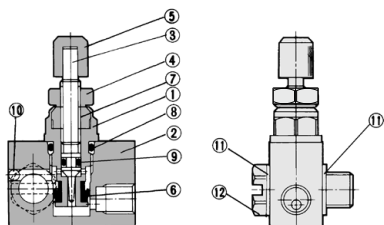
Be aware that all part numbers are for a meter-out type; there is no part number for a meter-in type.

Throttle Valve (Without Check Valve) **X214**

Ex.) AS2200-01-X214

Note) This product is a restrictor without a check valve (no control direction). Be aware that all part numbers are for a meter-out type; there is no part number for a meter-in type.

Construction: AS1400-M3



Component Parts

No.	Description	Material	Note
1	Body B	Brass	Electroless nickel plated
2	Body A	Brass	Electroless nickel plated
3	Needle	Brass	Electroless nickel plated
4	Lock nut	Steel ⁽¹⁾	Zinc chromated ⁽¹⁾
5	Handle	Brass	Electroless nickel plated
6	U seal	HNBR	
7	Needle guide	Brass	Electroless nickel plated
8	O-ring	NBR	
9	O-ring	NBR	
10	Steel ball	Chromium bearing steel	
11	Gasket	PVC	
12	Joint	Brass	Electroless nickel plated

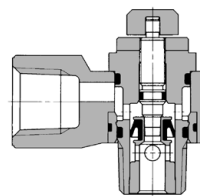
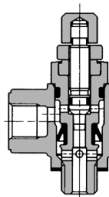
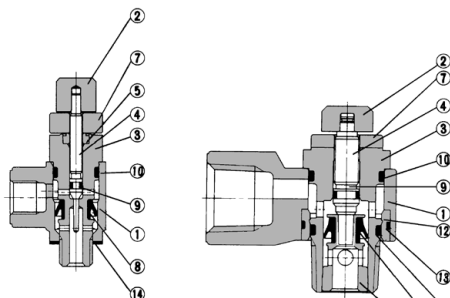
Note 1) The material and surface treatment of the lock nut option-J (round type) is brass and electroless nickel plating, respectively.

**Construction: AS1200-M3, AS12□0-M5,
AS22□0/32□0/42□0**

Meter-out type

AS1200-M3, M5, U10/32

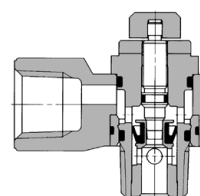
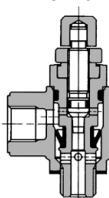
AS2200, 3200, 4200



Meter-in type

AS1210-M5. U10/32

AS2210, 3210, 4210



Component Parts

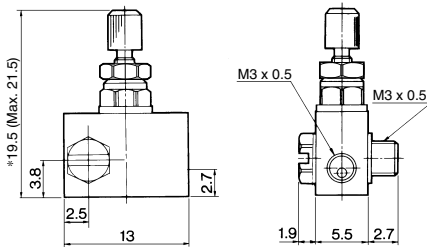
No.	Description	Material	Note	
1	Body A	Zinc alloy		
2	Handle	Brass	Electroless nickel plated	
3	Body B	Brass	Electroless nickel plated	
4	Needle	Brass	Electroless nickel plated	
5	Needle guide	Brass	Electroless nickel plated	
6	Seat ring	Brass	Electroless nickel plated ⁽¹⁾	
7	Lock nut	Steel ⁽²⁾	Meter-out	Zinc chromate plated ⁽³⁾
8	U seal	HNBR	Meter-in	Black zinc chromate plated
9	O-ring	NBR		
10	O-ring	NBR		
11	O-ring	NBR		
12	Bushing	PBT	01 to 04 type	
13	O-ring	NBR	01 to 04 type	
14	Gasket	PVC	M3 type	
		NBR, Stainless steel	M5, U10/32 type	

Note 1) For the AS42□0 type only, the electroless nickel plating specification is available for option N.

Note 2) The material of the lock nut option-J (round type) is brass. However, note that only the AS22□0 uses steel.

Note 3) The surface treatment of the lock nut option-J (round type) is electroless nickel plated.

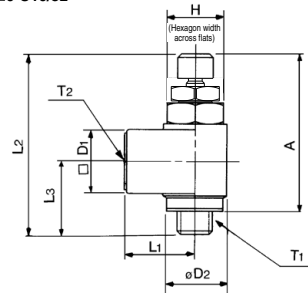
Dimensions: AS1400-M3



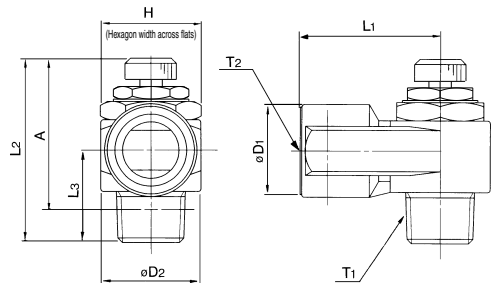
* Reference dimensions

Dimensions: AS-1200-M3, AS12□0-M5, AS22□0/32□0/42□0

AS1200-M3
AS12□0-M5
AS12□0-U10/32



AS22□0, 32□0, 42□0



Dimensions

Model	T1	T2	H (1)	L1	L2		L3	D1	D2	A (2)	
					Max.	Min.				Max.	Min.
AS1200-M3	M3 x 0.5	M3 x 0.5	4.5	6.6	23.5	21.5	8	5	5	20.5	18.5
AS12□0-M5	M5 x 0.8	M5 x 0.8	8	10	28.3	25.5	10.3	9	9	25	22.2
AS12□0-U10/32	10-32UNF	10-32UNF									
AS22□0-01	1/8	1/8	12 (12.7)	18	35.5	30.5	13.2	14.3	14.6	32.4	27.4
AS22□0-02	1/4	1/4	17 (17.5)	27.2	40.3	35.3	17.5	18	19.5	34.8	29.8
AS32□0-03	3/8	3/8	19	30	45.8	40.8	19.7	22.5	24.3	40.6	35.6
AS42□0-04	1/2	1/2	24 (23.8)	38.5	54.7	49.7	25.8	27.5	28.5	47.4	42.4

Note 1) () are the dimensions of "NPT" thread.

Note 2) Reference thread dimensions after installation.

Note 3) L2: Reference dimensions